

## **PBQ02 Lecture Objectives & Prerequisites**

### **Key Takeaways**

The coverage would include the following topics

- A quick recap from the last lecture
  - Primitive and advanced datatypes
  - Control structures
  - Functions in Python
- Libraries/Packages in Python
  - Understanding the hierarchy
  - Importing and working with various modules
- The NumPy Library
  - Understanding the NumPy ndarray data structure
  - Vectorization and how NumPy arrays are different from Python lists
  - Use-cases of various methods for NumPy arrays
  - Using mathematical functions from the NumPy library
- The pandas Library
  - Understanding pandas Series and DataFrames
  - Series creation and manipulation
  - Use-cases of various methods for Series
  - DataFrame creation and manipulation
  - Reading and writing csv/excel data
  - Use-cases of various methods for DataFrames
- Some commonly done calculations in quantitative trading
  - Percentage change, rolling, shift and other methods
  - Dealing with missing/NaN data
  - The groupby function

The below links take you to the official documentation of NumPy and pandas. These are excellent sources to understand these libraries.

NumPy: <https://numpy.org/doc/stable/>

Pandas: <https://pandas.pydata.org/pandas-docs/stable/10min.html>

### **Pre-requisites**

- PBQ-01 Lecture Recording

### **Downloadable Files**

- PBQ-02-Files.zip

### **Recommended time for the session & related coursework - 10 hours**

**Additional Read:**

**Blog:** [Basic Operations On Stock Data Using Python](#)

**Note:**

The lecture is conducted in a jupyter notebook environment. In case you do not have it installed on your local machine, you can run your jupyter notebook in the cloud without installation using:

- >Google (Colab)
- >Github
- >Binder
- >Kaggle Kernels
- >Microsoft Azure Notebooks